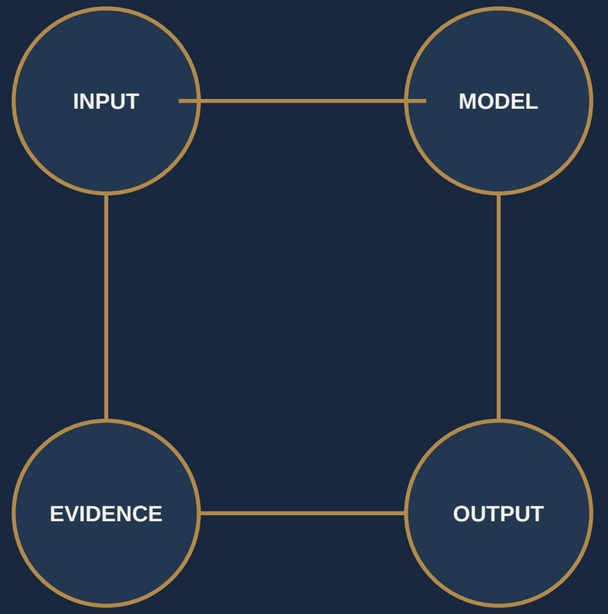
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Natural Language Processing

AI.105 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define text processing.
- Apply key methods including tokenization.
- Build or analyze a practical system involving stemming.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Text processing

02

METHODS

Tokenization

03

SYSTEMS

Stemming

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Text processing

Text processing

02 · Tokenization

Tokenization

03 · Stemming

Stemming

04 · Lemmatization

Lemmatization

05 · Language models

Language models

06 · Embeddings

Embeddings

07 · Sentiment analysis

Sentiment analysis

08 · Text classification

Text classification

Core ideas, made explicit

AI.105 • Distinguish definitions, assumptions and evidence

- Text processing
- Tokenization
- Stemming
- Lemmatization
- Language models

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.105 • Distinguish definitions, assumptions and evidence

- Embeddings
- Sentiment analysis
- Text classification
- Sequence models
- Transformers

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Text processing

INPUTS
Tokenization

METHOD
Stemming

OUTPUT
Lemmatization

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving text processing.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using text processing.
- Implement or simulate a baseline using tokenization.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.